WORKED SOLUTION

STEP III 1996, Question 11

Mechanics • Small Oscillations

STEP III 1996, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q11

The key idea

Two pieces of circle geometry do the work. AC is a diameter, so the angle ABC is a right angle. That gives both string lengths, and so both tensions, in terms of θ. Also, the angle at the centre is twice the angle at the circumference, so B has turned through 2θ about the centre. Its arc length from C is therefore $2a\theta$, and Newton's second law along the tangent gives the equation of motion. For the oscillation, linearise about the equilibrium.

A warning before we start: the printed equation has $mg\sin\theta$ where it should have $mg\sin 2\theta$. The derivation below shows why, and only the corrected equation gives the stated $5\lambda = 24mg$ and the stated period.

Solution

The equation of motion

Model. The ring B is a particle of mass m on a smooth wire, so the wire exerts only a normal reaction R along the radius. The strings are light and elastic and obey Hooke's law: a string of natural length a stretched to length ℓ has tension $\lambda(\ell - a)/a$.

Geometry. Since AC is a diameter (length $2a$), the angle in a semicircle gives $\angle ABC = 90^\circ$. In the right-angled triangle ABC,

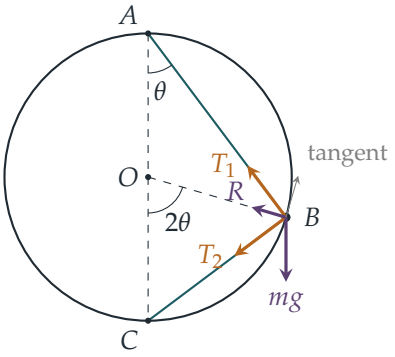
$$AB = 2a\cos\theta, \qquad CB = 2a\sin\theta.$$

The inscribed angle $\angle CAB = \theta$ stands on the arc CB, so the angle at the centre is $\angle COB = 2\theta$. The arc length from C to B is $s = 2a\theta$, so the acceleration of B along the wire, in the direction of increasing θ (towards A), is $\ddot{s} = 2a\ddot{\theta}$.

Tensions. The tensions in BA and BC are

$$T_1 = \lambda\,\frac{2a\cos\theta - a}{a} = \lambda(2\cos\theta - 1), \qquad T_2 = \lambda(2\sin\theta - 1).$$

Both strings are taut exactly when $\cos\theta > \frac{1}{2}$ and $\sin\theta > \frac{1}{2}$, that is, when $\frac{\pi}{6} < \theta < \frac{\pi}{3}$.



Resolving along the tangent at B. The tangent is perpendicular to OB, and R has no component along it.

- Triangle OAB is isosceles ($OA = OB = a$), so $\angle OBA = \angle OAB = \theta$. Hence BA makes an angle $90° - \theta$ with the tangent, and T_1 contributes $T_1 \sin\theta$ towards A.
- Triangle OCB is isosceles, and $\angle OCB = \angle ACB = 90° - \theta$, so $\angle OBC = 90° - \theta$. Hence BC makes an angle θ with the tangent, and T_2 contributes $T_2 \cos\theta$ away from A.
- OB makes an angle 2θ with the downward vertical, so the tangent makes an angle 2θ with the horizontal. Gravity contributes $mg \sin 2\theta$ away from A.

Newton's second law along the tangent gives

$$\begin{aligned} 2ma\ddot{\theta} &= \lambda(2\cos\theta - 1)\sin\theta - \lambda(2\sin\theta - 1)\cos\theta - mg\sin 2\theta \\ &= \lambda(\cos\theta - \sin\theta) - mg\sin 2\theta, \end{aligned}$$

since the $2\lambda \sin\theta\cos\theta$ terms cancel. This is the required equation, except that the gravity term is $mg\sin 2\theta$, not the printed $mg\sin\theta$.

Ask yourself Is the printed $mg\sin\theta$ perhaps right after all?

No. At $\theta = \frac{\pi}{4}$ the ring is level with O, the tangent is vertical, and gravity's full weight mg acts along the wire. That gives $mg\sin 2\theta = mg$, but $mg\sin\theta = mg/\sqrt{2}$. The next part settles it too: the printed equation would give $\lambda = 3mg$ at equilibrium, not $\frac{24}{5}mg$.

The equilibrium

With $\sin\theta = \frac{3}{5}$ we have $\cos\theta = \frac{4}{5}$ (taking θ acute), and θ lies in the taut range because $\frac{3}{5} > \frac{1}{2}$ and $\frac{4}{5} > \frac{1}{2}$. Setting $\ddot{\theta} = 0$:

$$\lambda\left(\tfrac{4}{5} - \tfrac{3}{5}\right) = mg \cdot 2 \cdot \tfrac{3}{5} \cdot \tfrac{4}{5} \quad\Longrightarrow\quad \frac{\lambda}{5} = \frac{24mg}{25} \quad\Longrightarrow\quad 5\lambda = 24mg,$$

as required.

Small oscillations

Write $f(\theta) = \lambda(\cos\theta - \sin\theta) - mg\sin 2\theta$, so that $2ma\ddot{\theta} = f(\theta)$ and $f(\theta_0) = 0$ at the equilibrium θ_0. Put $\theta = \theta_0 + \varepsilon$ with ε small. To first order, $f(\theta_0 + \varepsilon) \approx f'(\theta_0)\,\varepsilon$, where

$$\begin{aligned} f'(\theta_0) &= -\lambda(\sin\theta_0 + \cos\theta_0) - 2mg\cos 2\theta_0 \\ &= -\tfrac{24}{5}mg\cdot\tfrac{7}{5} - 2mg\left(\tfrac{16}{25} - \tfrac{9}{25}\right) = -\tfrac{168}{25}mg - \tfrac{14}{25}mg = -\tfrac{182}{25}mg. \end{aligned}$$

So

$$2ma\ddot{\varepsilon} = -\tfrac{182}{25}mg\,\varepsilon \quad\Longrightarrow\quad \ddot{\varepsilon} = -\frac{91g}{25a}\,\varepsilon.$$

This is simple harmonic motion with $\omega^2 = \dfrac{91g}{25a}$. Because $f'(\theta_0) < 0$, the equilibrium is stable. The displacement is slight, so the strings stay taut (θ stays near θ_0, well inside $(\tfrac{\pi}{6}, \tfrac{\pi}{3})$). The period is approximately

$$\frac{2\pi}{\omega} = 2\pi\cdot 5\sqrt{\frac{a}{91g}} = 10\pi\sqrt{\frac{a}{91g}}.$$

Checks

- At $\theta = \frac{\pi}{4}$ the strings are equally stretched, so their tangential effects cancel ($\cos\theta - \sin\theta = 0$), and the whole weight acts along the vertical tangent. Our equation gives $2ma\ddot{\theta} = -mg$, which is right.
- With $a = g = m = 1$ and $\lambda = 4.8$, we computed the tangential force from position vectors. It vanishes at $\sin\theta = 0.6$. Stepping the motion numerically from a small displacement gave a period of 3.29328, matching $10\pi/\sqrt{91} \approx 3.29328$.

Common slips

- Taking the arc length as $a\theta$. The ring turns through 2θ about the centre, not θ, so its tangential acceleration is $2a\ddot{\theta}$. This is where the 2 on the left comes from.
- Resolving along the chord directions instead of the tangent. The reaction is unknown, so resolve along the tangent to eliminate it.

Beyond STEP

Linearising about an equilibrium, $\ddot{x} = f(x) \approx f'(x_0)(x - x_0)$, is the first step in the stability analysis of nonlinear systems. It gives stability when $f'(x_0) < 0$ and simple harmonic motion with $\omega^2 = -f'(x_0)$. At university this extends to several degrees of freedom through normal modes and the eigenvalues of a matrix. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to read more.

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